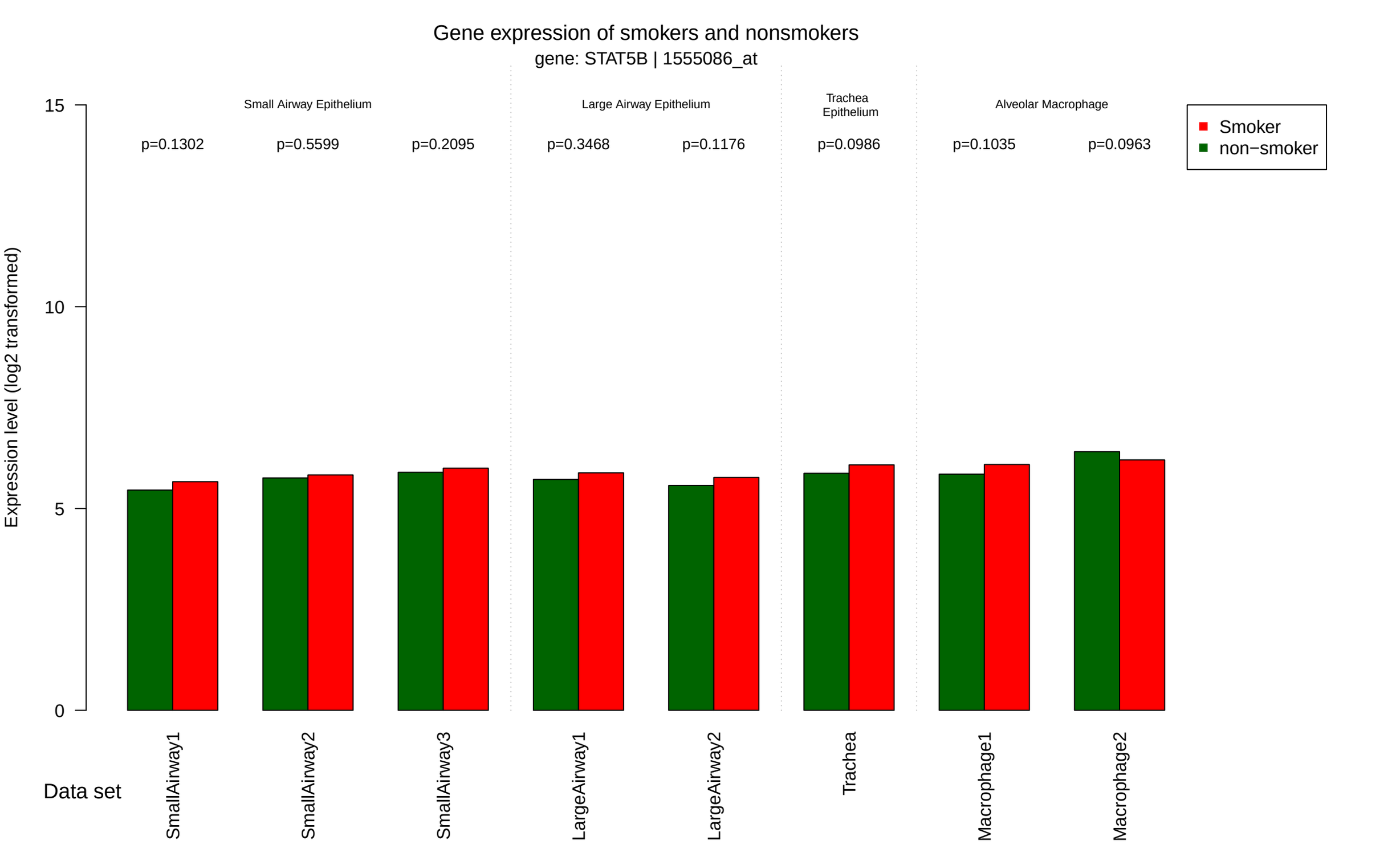
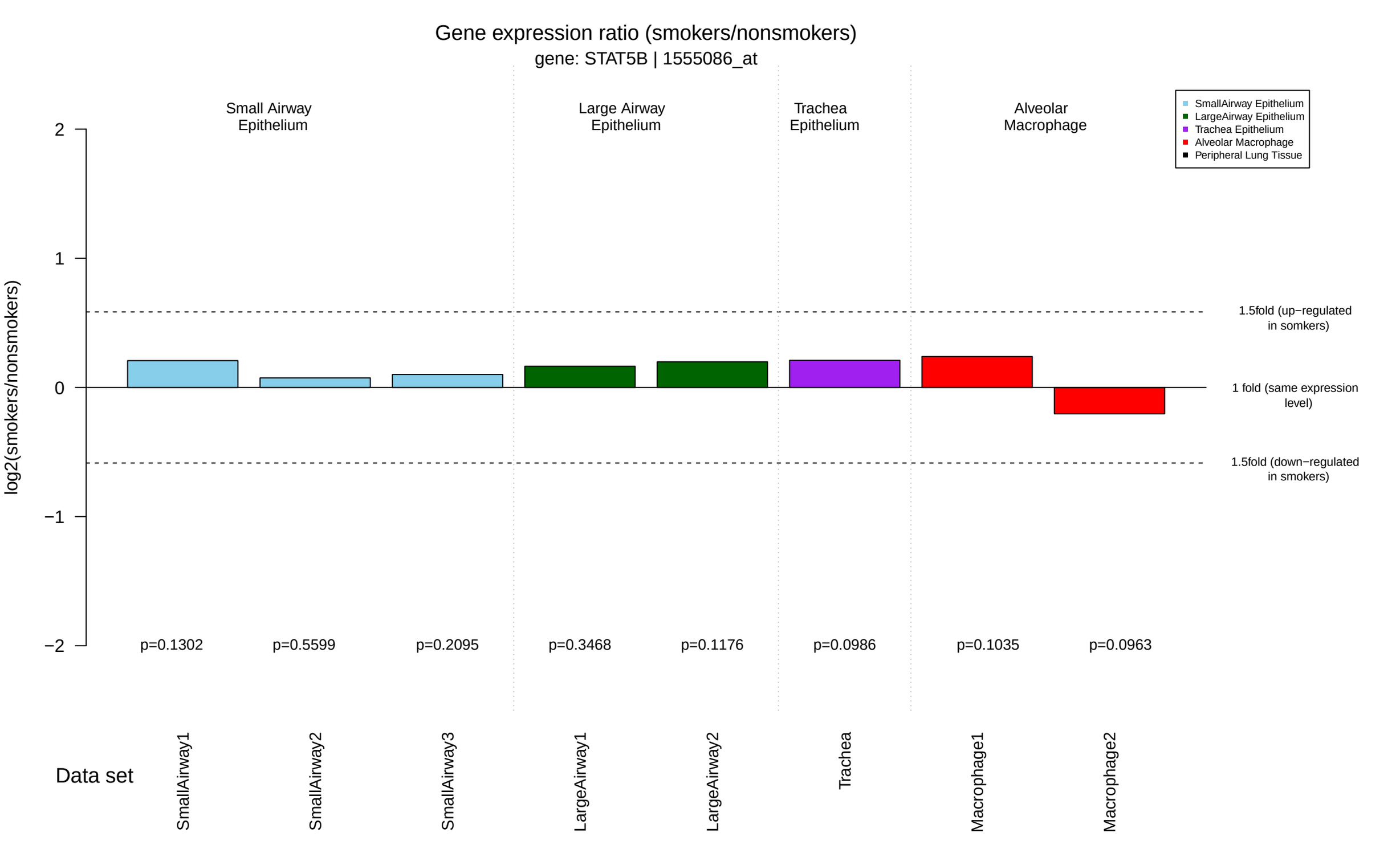
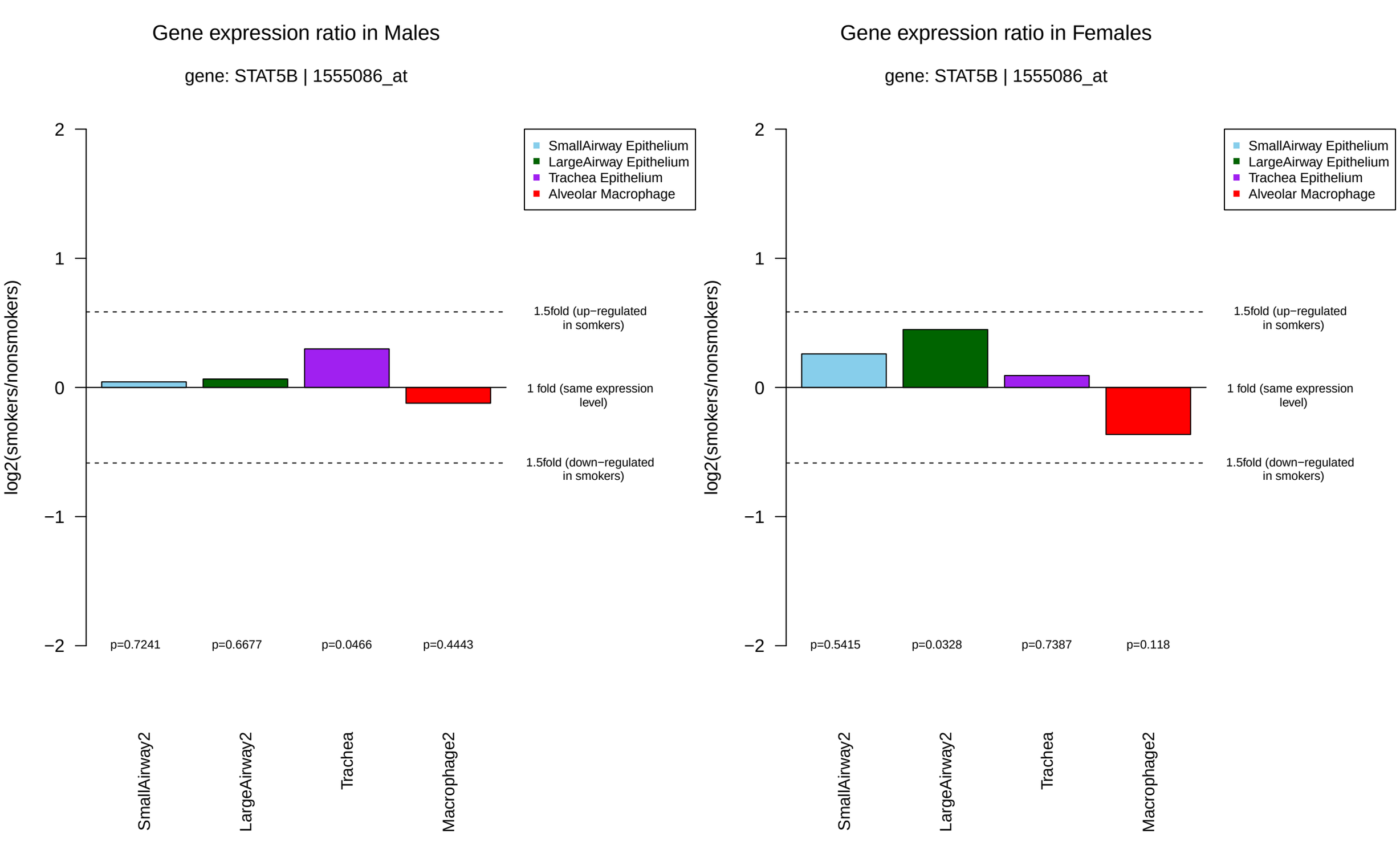
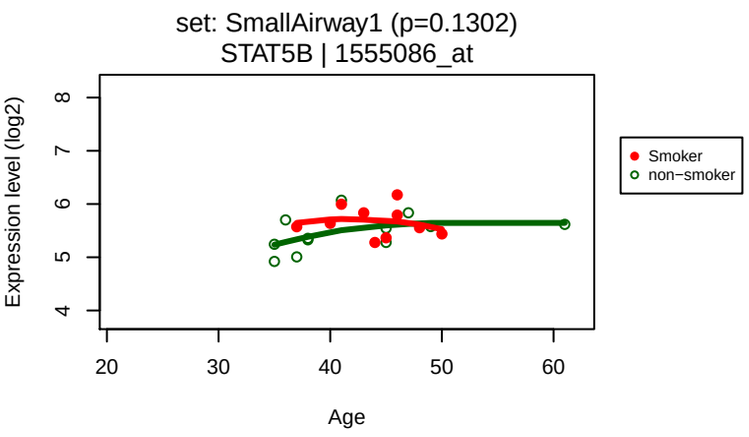
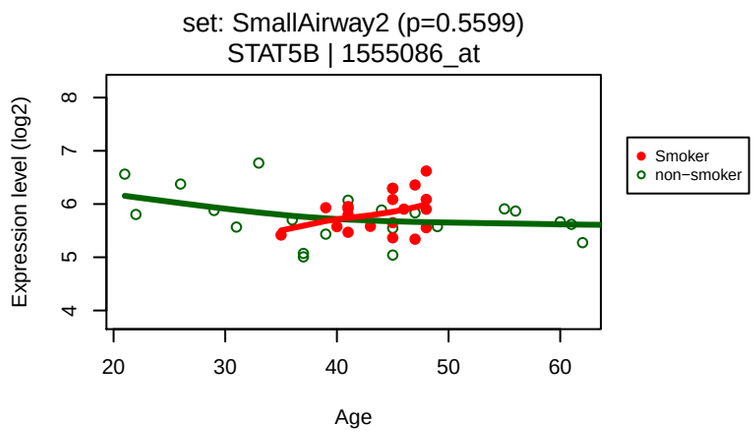
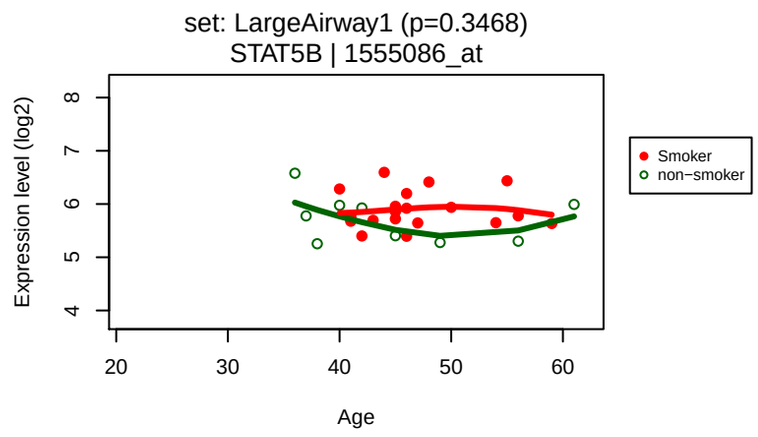

Gene expression of smokers and nonsmokers
gene: STAT5B | 1555086_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1302
p=0.5599
p=0.2095
p=0.3468
p=0.1176
p=0.0986
p=0.1035
p=0.0963
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

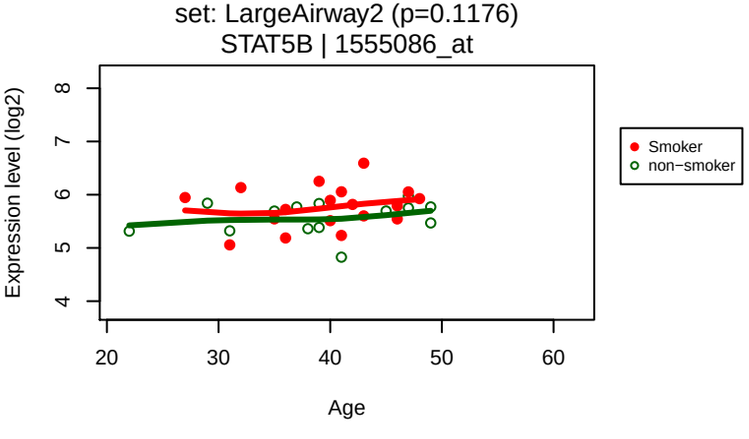

Gene expression ratio (smokers/nonsmokers)
gene: STAT5B | 1555086_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1302
p=0.5599
p=0.2095
p=0.3468
p=0.1176
p=0.0986
p=0.1035
p=0.0963
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

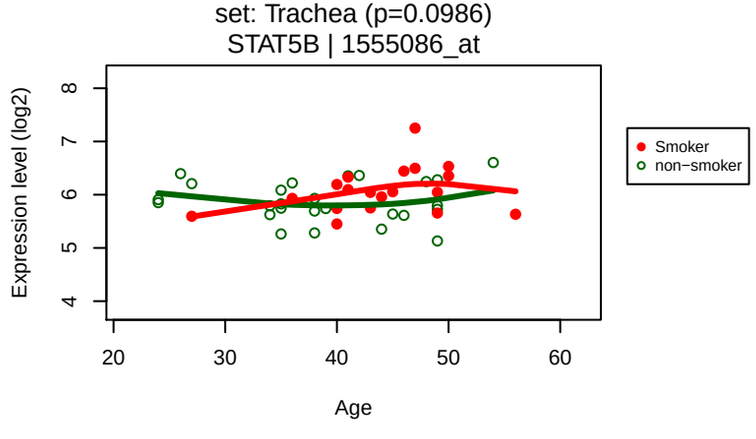

Gene expression ratio in Males
gene: STAT5B | 1555086_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7241
p=0.6677
p=0.0466
p=0.4443
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: STAT5B | 1555086_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5415
p=0.0328
p=0.7387
p=0.118
SmallAirway2
LargeAirway2
Trachea
Macrophage2

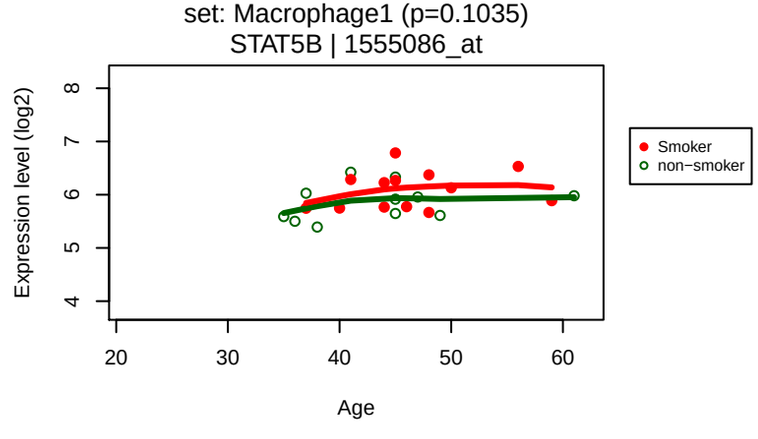

set: SmallAirway1 (p=0.1302)
STAT5B | 1555086_at
Expression level (log2)
Age
Smoker
non−smoker

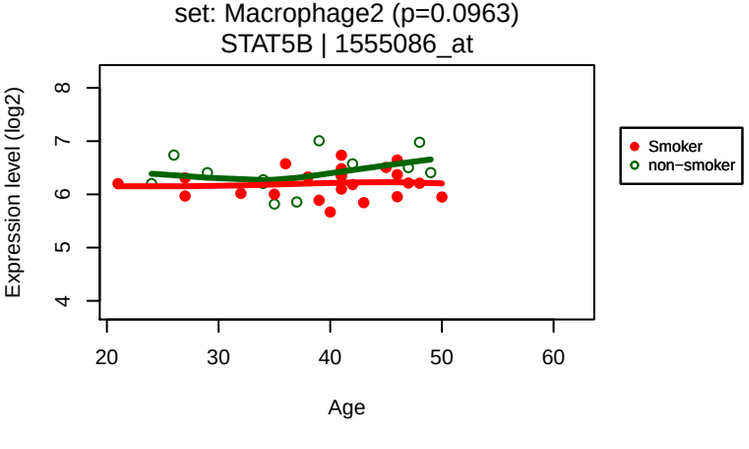

set: SmallAirway2 (p=0.5599)
STAT5B | 1555086_at
Expression level (log2)
Age
Smoker
non−smoker

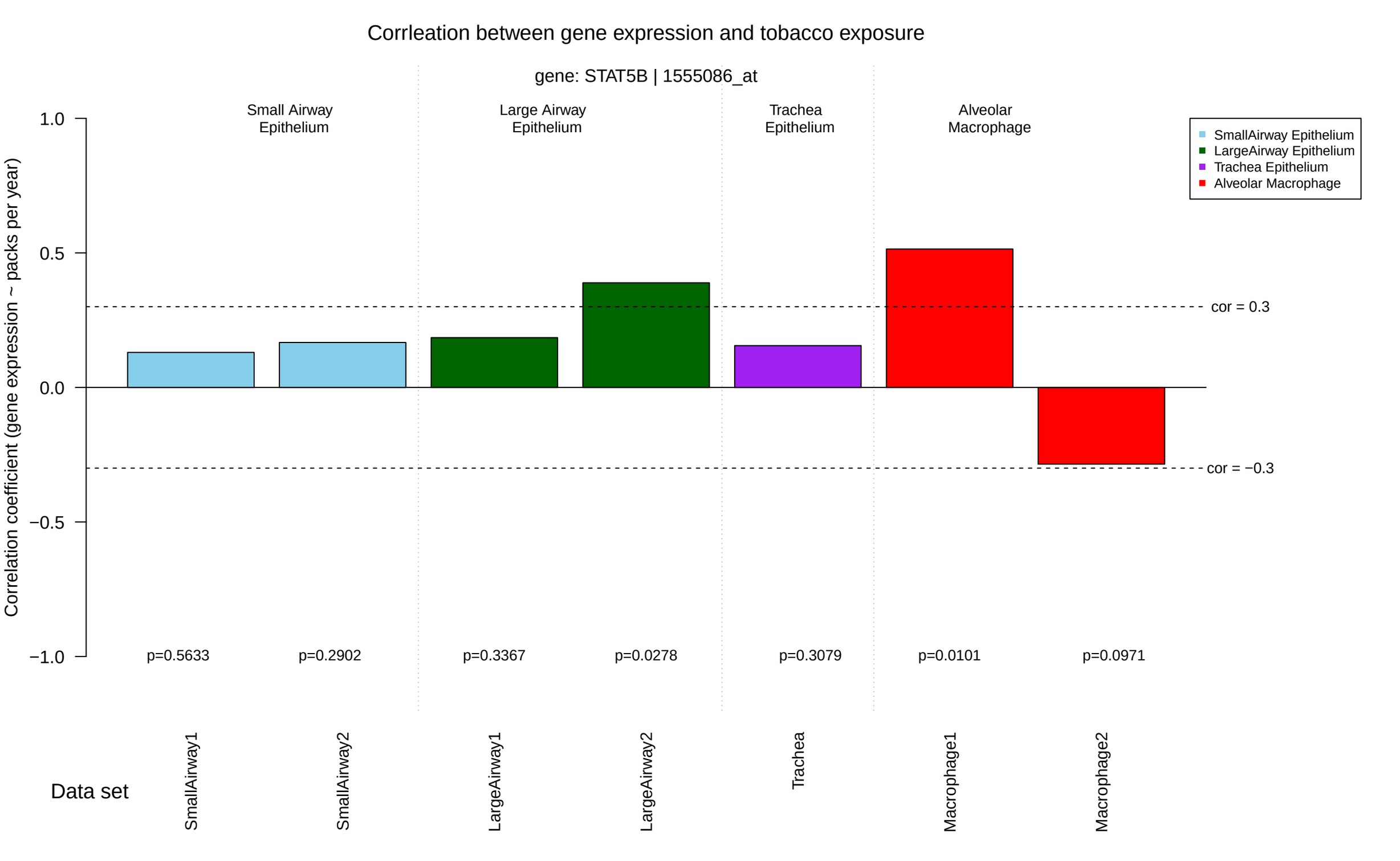

set: LargeAirway1 (p=0.3468)
STAT5B | 1555086_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1176)
STAT5B | 1555086_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0986)
STAT5B | 1555086_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1035)
STAT5B | 1555086_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0963)
STAT5B | 1555086_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: STAT5B | 1555086_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5633
p=0.2902
p=0.3367
p=0.0278
p=0.3079
p=0.0101
p=0.0971
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2